The Development of Metamotivational Knowledge:

How Children and Adults Understand Cognitive Reframing Shapes Motivation

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Background

- Knowledge about motivation supports effective regulation (Miele et al., 2024; Trautner & Pinquart, 2025).
- Cognitive reframing is a common strategy for regulating motivation, but evidence of children's understanding is mixed (Stockinger et al., 2025).
- Elementary schoolers show limited understanding of mental strategies for motivation regulation (Cooper & Corpus, 2009); emotion research suggests emerging sensitivity by ages 5–6 (Bamford & Lagattuta, 2012).

Research Question

How do children understand the role of cognitive reframing in shaping motivation?

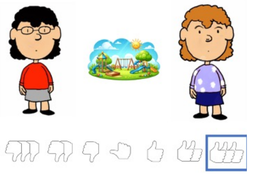
- Is this understanding reducible to valence-based reasoning or emotion understanding?

Study 1

3 (Situation Valence: positive, neutral, negative) × 2 (Reframing Valence: positive, negative)

Sherry and Beth really want to go outside to play right now (Sample Scenario - Positive Situation).

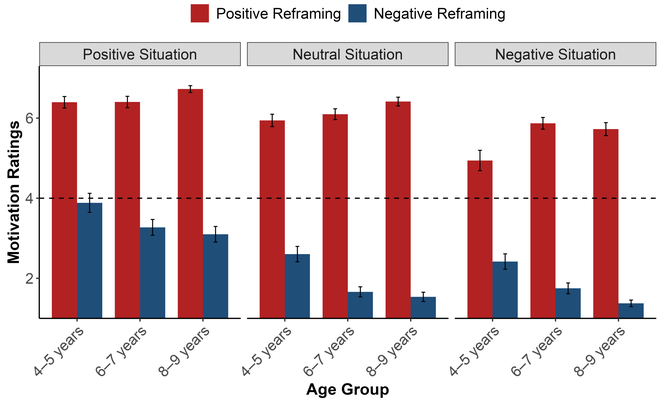
Positive Reframing:
"The weather is warm and nice. We'll see our friends and play lots of fun games!"



Negative Reframing:
"The weather is too hot. We will get sweaty and dirty from playing games with our friends!"

How much does XX want to go outside to play right now?

Results



- N = 129 (~40 per age group);
- Positive reframing > negative reframing (ps < .001, ds = 1.68 to 3.27);
- 4–5-year-olds < older age groups in difference scores (ps < .001, ds = -. 75 to -.56)

Study 2

2 (Situation Valence: positive, negative) × 2 (Thought Type: reframing, distraction) × 2 (Context: motivation, emotion) *(scenarios adapted from Bamford & Lagattuta, 2012)*

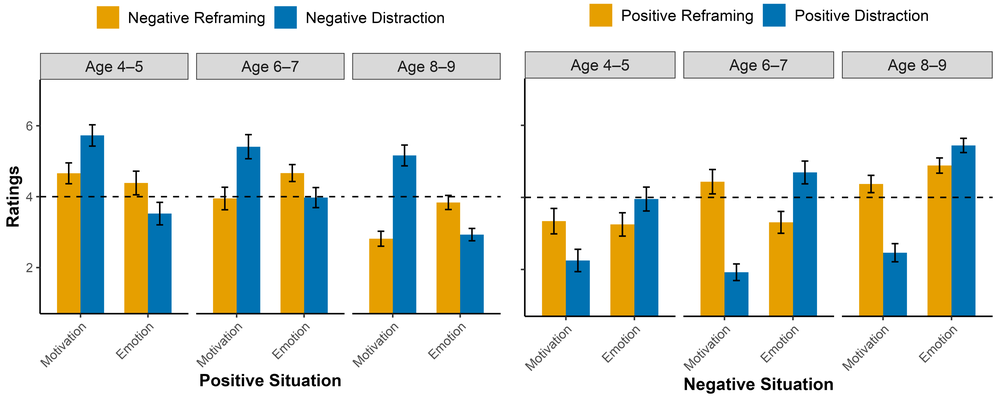
Positive Situation – Emotion (Example)
Amy and Erica felt very good when they first got the puppy.

- **Negative Reframing:** *"I got a new pet! It will be a lot of work for me to walk him and clean up after him every day!"*
- **Negative Distraction:** *"I got so much homework today. It will be a lot of work for me to finish everything and get all the questions right."*

Negative Situation – Motivation (Example)
Scott and Michae really did not want to clean up the mess when it happened.

- **Positive Reframing:** *"At least it didn't mess up our homework, and we can use our new sponges to clean it up!"*
- **Positive Distraction:** *"I'm wearing a very comfortable, lovely pair of shoes today. And they're brand new! I can wear them to go out later and play with my friends."*

Results



- N = 126 (~40 per age group);
- Reframing > distraction for motivation (ps < .003; ds = -.70 to 1.54 [positive situation], 0.64 to 1.47 [negative situation]), but similar for emotion;
- No reliable age group differences in differentiation between reframing and distraction.

Summary

- Even **young children recognized that reframing can shape motivation**, and this understanding goes beyond valence-based or emotion-based reasoning.
- **Older children attributed stronger effects** to reframing thoughts on motivation than younger children, particularly 4- to 5-year-olds.

Reference

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